

RISK INDICATORS FOR FRAUD

A. High-value transactions

Which users suddenly transact unusually high amounts?

SQL query

```
1 SELECT user_id, country,  
2 round (avg (amount),0) as avg_amt,  
3 round (max (amount),0) as max_amt,  
4 round (max (amount)/((SUM(amount)- MAX(amount)) / (COUNT(*) - 1)), 0) as risk_ratio  
5 FROM creditcard01  
6 group by user_id, country  
7 HAVING COUNT(*) > 2  
8 order by risk_ratio DESC;  
9
```

	user_id	country	avg_amt	max_amt	risk_ratio
1	CC_8408	UK	1750.0	4881.0	26.0
2	CC_9751	India	1420.0	3575.0	10.0
3	CC_1325	India	2642.0	4792.0	3.0
4	CC_2137	India	1800.0	3144.0	3.0
5	CC_3892	UK	2146.0	4026.0	3.0
6	CC_6517	Singapore	2287.0	3949.0	3.0
7	CC_9017	Australia	1501.0	2785.0	3.0

Execution finished without errors.
Result: 25 rows returned in 10ms

Observation - Behavioural risk analysis identified 25 users whose maximum transaction values significantly exceeded their normal spending patterns. The highest anomaly ratios reached 26x in the UK and 10x in India, indicating potentially suspicious transaction spikes requiring further investigation.

Why It Matters - Large spikes may indicate suspicious or unauthorized activity.

Risk Level - High

Recommended Action - Flag for manual review.

B. High-frequency transactions

Which users perform excessive transactions in short periods?

SQL query

```

1  SELECT user_id,
2  country,
3  substr(timestamp, 1, 13) as txn_hour,
4  count(*) as transaction_count,
5  round(sum(amount),0) as total_amount
6  FROM creditcard01
7  group by user_id, country, txn_hour
8  HAVING COUNT(*) > 1
9  order by transaction_count DESC;
10

```

user_id	country	txn_hour	transaction_count	total_amount
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Execution finished without errors.
Result: 0 rows returned in 10ms

Observation - No users were identified with excessive transaction frequency within the same hourly period, indicating the absence of significant short-duration transaction bursts or unusually high transaction velocity patterns in the data.

Why It Matters - Rapid transaction patterns may indicate automated transaction attempts, abnormal user activity or potential fraud-related transactions.

Risk Level - Low

Recommended Action - Continue monitoring transaction patterns and implement alert thresholds for unusually high transaction frequency within short time periods.

C. Threshold avoidance

Are users repeatedly transacting just below review limits?

SQL query

```

1  SELECT user_id,
2         country,
3         round (avg(amount),0) as avg_amt,
4         round (max(amount),0) as max_amt,
5         count(*) as trans_count
6  FROM creditcard01
7  WHERE amount BETWEEN 3000 AND 3999
8  GROUP by user_id, country
9  HAVING count(*) > 2
10 ORDER by trans_count DESC;
11

```

	user_id	country	avg_amt	max_amt	trans_count
1	CC_9246	UK	3395.0	3763.0	3

Execution finished without errors.
Result: 1 rows returned in 12ms

Observation - User CC_9246 from the UK performed 3 repeated transactions within the 3000–3999 range, indicating potential attempts to avoid transaction review thresholds.

Why It Matters - Repeated transactions just below review thresholds may indicate attempts to bypass automated fraud detection.

Risk Level - Medium

Recommended Action - Monitor repeated near-threshold transaction patterns and trigger additional verification checks for recurring activity below review limits.

D. Geographic inconsistency

Are there unusual country/IP changes?

SQL query

```

1  SELECT user_id,
2  COUNT(DISTINCT country) AS country_count,
3  COUNT(DISTINCT ip_address) AS ip_count
4  FROM creditcard01
5  GROUP BY user_id
6  HAVING country_count > 5
7         OR ip_count > 5
8  ORDER BY country_count DESC, ip_count DESC;
9

```

	user_id	country_count	ip_count
1	CC_6515	5	7
2	CC_2642	5	6
3	CC_9535	5	6
4	CC_2415	4	6
5	CC_2137	3	6
6	CC_9864	3	6

Execution finished without errors.
Result: 6 rows returned in 18ms

Observation - Six users exhibited transactions across multiple countries and IP addresses, indicating potential geographic inconsistencies in transaction behaviour.

Why It Matters - Frequent changes in transaction locations or IP addresses may indicate account compromise, VPN/proxy usage, or unauthorized access.

Risk Level - Medium

Recommended Action - Trigger additional authentication and review transactions originating from multiple countries or IP addresses within a short period.

E. Behavioural deviation

Does current behaviour differ from historical activity?

SQL query

```

1  SELECT user_id, country,
2  round( Avg(amount), 0) AS avg_amt,
3  round( max(amount), 0) AS highest_amt,
4  round(max(amount)/Avg(amount),2) as deviation_ratio
5  FROM creditcard01
6  GROUP by user_id, country
7  HAVING deviation_ratio > 2
8  ORDER by deviation_ratio DESC;
9

```

	user_id	country	avg_amt	highest_amt	deviation_ratio
1	CC_8408	UK	1750.0	4881.0	2.79
2	CC_9751	India	1420.0	3575.0	2.52

Execution finished without errors.
 Result: 2 rows returned in 17ms
 At line 1:

Observation - Two users exhibited transaction amounts more than twice their average spending, indicating a significant deviation from their normal spending behaviour.

Why It Matters - Transactions that differ substantially from a user's typical spending pattern may indicate unauthorized account activity or potential fraud.

Risk Level - Medium

Recommended Action - Flag transactions with high behavioural deviation for additional verification and monitor users for further unusual activity.